

Assignment: Frequent Pattern Mining with Apriori

Dataset

Use the Groceries Dataset (Groceries_dataset.csv)

Load and Explore the Dataset

1. Load the CSV file using pandas.
2. Display the first 10 rows.
3. Identify the columns representing transaction ID and items.
4. Report the number of rows and columns in the dataset.

Question: **What does a transaction represent in this dataset?**

Create Transactions

1. Convert the dataset into a list of transactions.
2. Each transaction should contain a list of items purchased together.
3. Convert this into a one-hot encoded basket matrix.
4. Display the shape of the basket matrix.

Question: **What does each row and column represent?**

Apply Apriori Algorithm

Run Apriori with two different support thresholds.

Example thresholds:

`min_support = 0.01`

`min_support = 0.02`

Tasks:

1. List the top 10 frequent itemsets.
2. Report the total number of frequent itemsets discovered.
3. Identify the largest itemset length discovered.

Association Rules

Generate association rules from the frequent itemsets.

Show the following columns:

Antecedents

Consequents

Support

Confidence

Lift

Tasks:

1. Display the top 10 rules with the highest lift.
2. Filter rules where confidence ≥ 0.50 and lift ≥ 1.2 .

Manual Calculations (Important)

For the following rule:

Bread $\rightarrow$ Milk

Assume the following values:

Total transactions = 100

Transactions containing Bread = 40

Transactions containing Milk = 50

Transactions containing both Bread and Milk = 30

Calculate:

1. Support
2. Confidence
3. Lift

Show all formulas and calculations.

Data Visualization

Create the following visualizations using Python (matplotlib):

1. Bar Chart: Top 15 most frequent items.
2. Scatter Plot: Support vs Confidence.
3. Histogram: Distribution of Lift values.

Interpretation Questions:

1. Which items appear most frequently?
2. Where do the strongest rules appear on the scatter plot?

3. Why is Lift useful when evaluating association rules?

Pivot Table Analysis

Before applying Apriori, explore the dataset using a pivot table.

Task 1:

Create a pivot table showing the number of times each item appears in transactions.

Rows: itemDescription

Values: count of transactions

Task 2:

Identify the top 10 most frequently purchased items.

Sort them in descending order.

Task 3:

Create a bar chart showing the top 10 most frequently purchased items.

Interpretation Questions:

1. Which item appears most frequently in the dataset?
2. Why is it useful to identify high-frequency items before applying Apriori?
3. Do the most frequent items necessarily produce the strongest association rules? Explain.